ATmega32U4 handheld enclosure: print and assembly notes

For Haris · FS10 power supply and monitor · closed size 89 × 142.5 × 33 mm · STEP files (exact solids; your slicer meshes them)

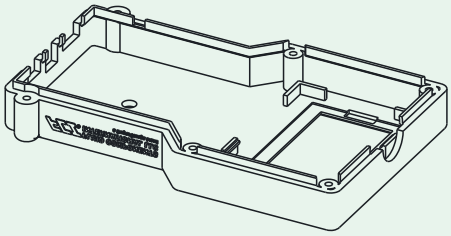
FILES

File	Print?	Size (mm)	Notes
`ATmega32U4-Handheld-Front_Shell.step`	**yes**	89.7 × 143.1 × 20.0	display, LED window, header pockets, logo
`ATmega32U4-Handheld-Back_Shell.step`	**yes**	89.7 × 143.1 × 16.5	floor, board screw bosses, reset pinhole
`ATmega32U4-Handheld-Display_coupon.step`	optional, first	89 × 63 × 20	head of the front shell only: quick fit check of the window, clips and screw pilots in your plastic
`ATmega32U4-Handheld-Board_fitcheck.step`	optional	58 × 71 × 15	stand-in board (outline, holes, connectors) for a dry fit
`ATmega32U4-Handheld.step`	no	—	both shells assembled, for reference

ORIENTATION: NO SUPPORTS ON EITHER PART

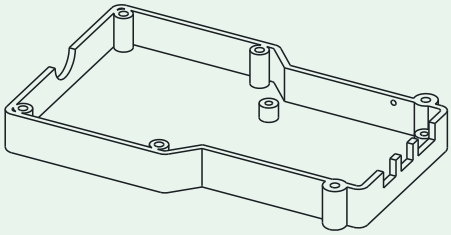
Front shell: face-down

outside face (window, bevel) on the bed; clips and rails point up



Back shell: floor-down

outside of the floor on the bed; bosses point up



SETTINGS (RED PETG / ABS / ASA)

Layer height	0.20 mm
Walls	**4** (screw bosses and pilots)
Top / bottom	5 layers each
Infill	30–40 %, gyroid
Supports	off
Brim	PETG: none. ABS/ASA: 5 mm, enclosed printer, low fan. The parts are 143 mm long and corners lift.
Scaling	100 %. Check the coupon before compensating for shrink.

DESIGNED FITS

- Halves: 0.3 mm clearance, locating lip on the front shell
- M3 self-tapper pilots **2.7 mm** (case and board); clearance holes 3.4 mm, head counterbore 6.2 mm
- Window pocket 64.4 × 40.4 × 0.6 mm for 0.5 mm polycarbonate
- Reset pinhole 2.0 mm, 4.0 mm above the board top; LED window 5.4 mm

HARDWARE

- 6 × M3 × 22–25 mm self-tapping, pan head ≤ 6 mm: case, from the back
- 4 × M3 × 6 mm self-tapping: board onto the printed bosses
- 0.5 mm polycarbonate, cut 64 × 40 mm, corners nicked 0.5 mm
- 2.42″ SSD1309 I2C OLED module, 70 × 48 mm board
- M12 8-pin panel connector, PG9 (Mencom MDC-8FR-PG9) and jam nut

ASSEMBLY

1. Front shell: drop the polycarbonate into its pocket, then snap the OLED into the two clips, glass to the window. Its wired 4-pin header sits on the left (seen from the front) over its pocket.
2. Back shell: screw the board onto the four bosses (M3 × 6). Drive gently; stop when snug.
3. Fit the M12 in the end wall. The hole is split by the seam; the jam nut sits inside.
4. Solder the OLED wires, close the halves on the lip, and fit the 6 screws from the back.
5. Check: USB, DC and the power switch sit flush in the bottom wall; a paper clip in the left-side pinhole clicks RESET.

> Tell Jesse about anything that binds, cracks or needs force. The model is parametric, so fits are quick to adjust.

Files exported 2026-10-08 from ATmega32U4-Handheld.FCStd (build_atmega32u4_handheld.py); each part checked as one valid, closed solid. The same files went to Ed at FCI for a resin print.